

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The Solution Requirements define the functional and non-functional requirements of the Smart Lender – Loan Approval Prediction System. These requirements describe the system's features, performance, security, usability, and operational standards to ensure accurate loan prediction and a reliable user experience.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>No user authentication is required as the application is a public prediction tool.</i>	Low
2	Authorization levels	<i>All users have the same access to enter loan details and receive predictions.</i>	Low

3	External interfaces	<i>Uses a Flask web interface and loads the trained ML model (smart_lender_model.pkl).</i>	High
4	Transactions processing	<i>Accepts user inputs, processes them, predicts loan status, and displays the result.</i>	High
5	Reporting	<i>Displays the loan prediction (Approved/Rejected) on the web page.</i>	Medium
6	Business rules, etc.	<i>Prediction is based on applicant details such as income, credit history, loan amount, and property area.</i>	High
7	Compliance to laws or regulations	<i>User data is processed only for prediction and is not stored permanently.</i>	Medium
8	Other	<i>Provides a simple and interactive interface for loan eligibility prediction.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>Generates loan predictions quickly after user submission.</i>	<i>Prediction in < 2 seconds</i>
2	Scalability	<i>Supports multiple prediction requests through the deployed web application.</i>	<i>Supports 100+ users/day</i>
3	Security & Data Privacy	<i>User input is processed securely and is not permanently stored.</i>	<i>No permanent storage of user data</i>
4	Reliability & Availability	<i>Provides consistent prediction results whenever the application is running.</i>	<i>99% application availability</i>
5	Usability & Accessibility	<i>Simple web interface with clear input fields and prediction output.</i>	<i>Easy to use with minimal user effort</i>
6	Other	<i>Portable and easy to deploy using Flask and Render.</i>	<i>Runs on Windows and cloud deployment</i>

Overall Description:-

The Smart Lender – Loan Approval Prediction System is a machine learning-based web application developed to simplify and improve the loan eligibility evaluation process. The system enables users to enter their personal and financial information through an easy-to-use web interface and instantly receive a prediction indicating whether their loan is likely to be approved or rejected. By utilizing historical loan data and a trained machine learning model, the application provides quick, reliable, and consistent predictions that can support preliminary loan assessment.

The functional requirements define the essential capabilities of the system, including accepting user input, validating the entered information, processing the data using the trained machine learning model, and displaying the prediction result through the web application. These requirements ensure that the application performs its intended functions accurately while providing a smooth and interactive experience for users. The business logic implemented within the application considers multiple factors such as applicant income, co-applicant income, credit history, loan amount, loan term, education, employment status, and property area to generate meaningful predictions.

The non-functional requirements focus on the quality attributes of the system, including performance, security, reliability, scalability, usability, and maintainability. The application is designed to generate predictions within a short response time, provide a simple and user-friendly interface, and maintain consistent performance under normal usage conditions. User data is processed only for prediction purposes and is not permanently stored, ensuring basic privacy during application usage. Furthermore, the application is lightweight, portable, and can be easily deployed using Flask and cloud platforms such as Render, making it accessible from different devices through a web browser.

Overall, these functional and non-functional requirements establish a strong foundation for the Smart Lender – Loan Approval Prediction System. They ensure that the application not only performs its intended tasks efficiently but also provides a reliable, secure, and user-friendly experience. By combining machine learning with a web-based interface, the system offers an effective solution for preliminary loan approval prediction while demonstrating the practical application of artificial intelligence in the banking and financial services domain.